SYLLABUS

IV Year B. Tech CSE – AIML - I Sem **L/T/P/C 3/-/-/-3**

(R20A6610) DEEP LEARNING

COURSE OBJECTIVES:

1. To understand the basic concepts and techniques of Deep Learning and the need of Deep Learningtechniques in real-world problems
2. To understand CNN algorithms and the way to evaluate performance of the CNN architectures.
3. To apply RNN and LSTM to learn, predict and classify the real-world problems in the paradigms of DeepLearning.
4. To understand, learn and design GANs for the selected problems.
5. To understand the concept of Auto-encoders and enhancing GANs using auto-encoders.

UNIT-I:

INTRODUCTION TO DEEP LEARNING: Historical Trends in Deep Learning, Why DL is Growing, Artificial Neural Network, Non-linear classification example using Neural Networks: XOR/XNOR, Single/Multiple Layer Perceptron, Feed Forward Network, Deep Feed- forward networks, Stochastic Gradient –Based learning, Hidden Units, Architecture Design, Back- Propagation.

UNIT-II:

CONVOLUTION NEURAL NETWORK (CNN): Introduction to CNNs and their applications in computer vision, CNN basic architecture, Activation functions-sigmoid, tanh, ReLU, Softmax layer, Types of pooling layers, Training of CNN in TensorFlow, various popular CNN architectures: VGG, Google Net, ResNet etc, Dropout, Normalization, Data augmentation

UNIT-III

RECURRENT NEURAL NETWORK (RNN): Introduction to RNNs and their applications in sequential data analysis, Back propagation through time (BPTT), Vanishing Gradient Problem, gradient clipping Long Short Term Memory (LSTM) Networks, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional RNNs.

UNIT- IV

GENERATIVE ADVERSARIAL NETWORKS (GANS): Generative models, Concept and principles of GANs, Architecture of GANs (generator and discriminator networks), Comparison between discriminative and generative models, Generative Adversarial Networks (GANs), Applications of GANs.

UNIT- V

AUTO-ENCODERS: Auto-encoders, Architecture and components of auto-encoders (encoder and decoder), Training an auto-encoder for data compression and reconstruction, Relationship between Autoencoders and GANs, Hybrid Models: Encoder-Decoder GANs.

TEXT BOOKS:

1. Deep Learning: An MIT Press Book by Ian Goodfellow and Yoshua Bengio Aaron Courville.
2. Michael Nielson, Neural Networks and Deep Learning, Determination Press,2015.
3. Satish Kumar, Neural networks: A classroom Approach, Tata McGraw-Hill Education, 2004.

REFERENCES:

1. Deep Learning with Python, Francois Chollet, Manning publications, 2018
2. Advanced Deep Learning with Keras, Rowel Atienza, PACKT Publications, 2018

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CO5: Understand the concept of Auto-encoders and enhancing GANs using auto-encoders.